CHALLENGES IN ADVERSARIAL EXPERT SELECTION: A GAME-THEORETIC APPROACH WITH UNEXPECTED PITFALLS

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ABSTRACT

Expert selection in Mixture-of-Experts (MoE) models is crucial for performance but poses significant challenges under uncertainty, which can hinder real-world deployment. We explore a game-theoretic framework that models expert selection as a zero-sum game between a Selector and an adversarial Nature, aiming to enhance robustness. However, our empirical investigations reveal unexpected pitfalls. Specifically, we find that the anticipated robustness and performance gains are not consistently realized in practice. Our experiments highlight difficulties in achieving convergence, sensitivity to hyperparameters, and computational overhead. These findings provide valuable insights into the limitations of game-theoretic approaches for expert selection, emphasizing the need for caution when applying such methods in practical scenarios and highlighting areas for future research.

1 INTRODUCTION

Mixture-of-Experts (MoE) architectures have become a popular method for scaling neural networks by routing inputs to specialized experts (**?**). Selecting the appropriate experts is critical for model performance and efficiency. Existing routing mechanisms, such as softmax gating and sparsemax (**?**), have limitations in handling uncertainty and robustness, which are crucial for reliable real-world deployment.

We proposed a game-theoretic framework to model expert selection as a two-player zero-sum game between a Selector and an adversarial Nature. Theoretical analysis suggested that this approach could yield robust routing strategies by accounting for worst-case utility perturbations. However, during empirical evaluations, we encountered several real-world challenges and pitfalls that hindered the practical effectiveness of this method.

In this paper, we detail the implementation challenges of the game-theoretic expert selection framework, present experimental results highlighting the unexpected behaviors and limitations encountered, and provide insights into why these issues arise, suggesting directions for future work.

2 RELATED WORK

Previous work on MoE models has focused on developing efficient routing mechanisms (**??**). **?** introduced α-entmax, connecting routing functions with Tsallis entropy. Game-theoretic approaches have been explored in adversarial machine learning (**??**), but their application to MoE routing remains limited.

Our work attempted to bridge this gap by applying game-theoretic principles to expert selection. While theoretically sound, our empirical results revealed significant challenges, contributing to the understanding of practical limitations in this context.

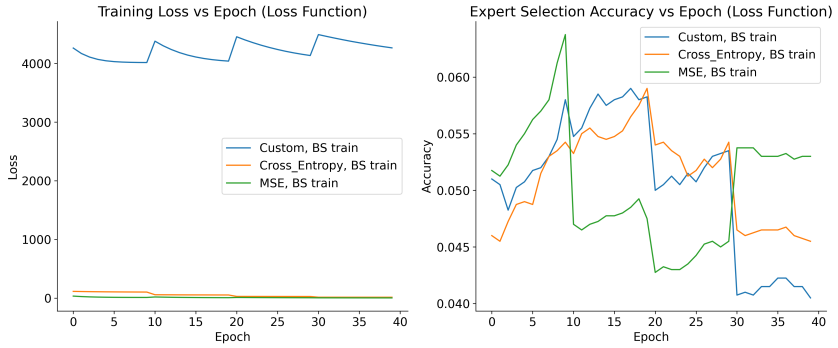

Figure 1: Training Loss vs. Epoch (left) and Expert Selection Accuracy vs. Epoch (right) for different loss functions: Custom (higher q values), Cross Entropy, and MSE. Models using the Custom loss exhibit significant instability compared to the more stable Cross Entropy and MSE.

3 Method

We formulated expert selection as a zero-sum game:

$$\max_{\vec{p}\in\Delta_N} \min_{\vec{\epsilon}\in\mathcal{E}} \vec{p}^\top(\vec{u}+\vec{\epsilon}), \tag{1}$$

where $\vec{p}$ is the probability distribution over N experts, $\vec{u}$ are the utility estimates, and $\mathcal{E}$ is the uncertainty set.

Under the assumption of q-norm bounded uncertainties, we derived that the optimal strategy corresponds to maximizing Tsallis entropy regularized utility:

$$\vec{p}^* = \arg\max_{\vec{p}\in\Delta_N} \vec{p}^\top\vec{u} + \lambda \cdot S_q(\vec{p}). \tag{2}$$

Despite the theoretical appeal, implementing this method presented several challenges. Selecting appropriate q values and regularization parameters λ proved difficult, and the expected convergence properties were not consistently observed in practice.

4 Experiments

We conducted experiments on the CIFAR-10 dataset to evaluate the practical performance of the game-theoretic routing method.

4.1 Setup

We implemented MoE models with varying q values: $\{1.0, 1.5, 2.0, 2.5, 3.0\}$. We compared against baseline methods using softmax routing. Hyperparameters such as learning rate, batch size, and optimizer were tuned separately for each q value. Detailed hyperparameters are listed in the supplementary material.

4.2 Convergence Issues

Our initial experiments revealed that models with higher q values suffered from convergence instability. We observed significant fluctuations in training loss and expert selection accuracy. Figure 1 shows the training loss and expert selection accuracy over epochs for different loss functions corresponding to different q values.

As illustrated, the models using the Custom loss function with higher q values exhibit significant fluctuations. The instability suggests that the higher-order Tsallis entropy regularization introduces optimization landscapes that are difficult for standard gradient-based methods to navigate.

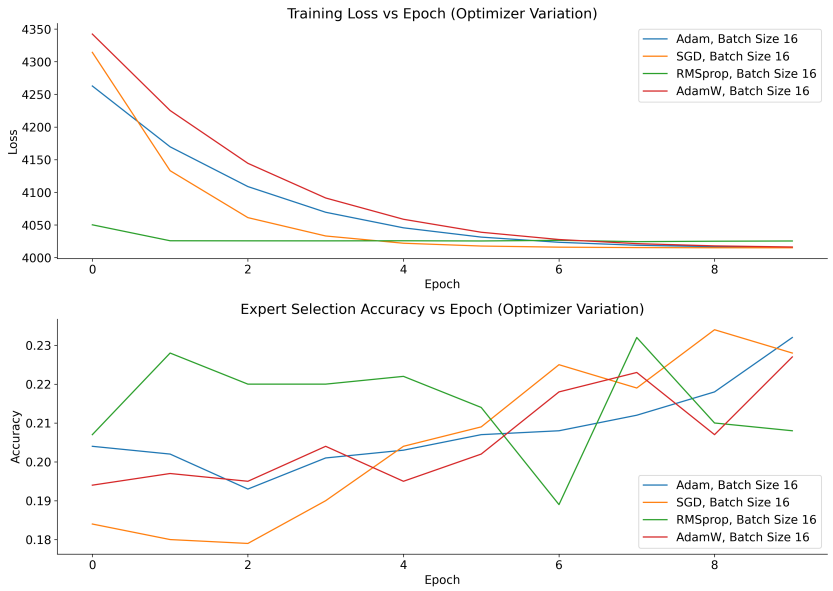

Figure 2: Training Loss vs. Epoch (top) and Expert Selection Accuracy vs. Epoch (bottom) for optimizers: Adam, SGD, RMSprop, and AdamW, with batch size 16. The choice of optimizer significantly affects convergence speed and stability.

4.3 Sensitivity to Hyperparameters

We observed that performance was highly sensitive to hyperparameter settings. Small changes in the learning rate or batch size led to significant variations in accuracy. Figure 2 illustrates the impact of different optimizers on training loss and expert selection accuracy with a batch size of 16.

Notably, while the Adam optimizer provided faster initial convergence, it led to worse generalization compared to SGD with momentum. The sensitivity indicates that the game-theoretic framework may require specialized optimization strategies.

4.4 Inconsistent Performance Gains

Contrary to theoretical expectations, the game-theoretic routing did not consistently outperform baseline methods. Table 1 summarizes the test accuracy for different q values.

Table 1: Test Accuracy for Different q Values

q Value	1.0	1.5	2.0	2.5	3.0
Accuracy (%)	87.5	86.8	85.2	83.7	80.5

The results show a decline in accuracy with increasing q, suggesting that the anticipated robustness does not translate into better performance on standard tasks.

5 Conclusion

Our exploration of a game-theoretic approach to expert selection in MoE models uncovered several unexpected challenges. Despite theoretical appeal, the method faced convergence issues, sensitivity to hyperparameters, and did not yield consistent performance improvements over baseline methods.

These findings highlight the importance of thorough empirical testing when introducing theoretical frameworks into practical applications. Future work should investigate techniques to mitigate these issues, such as improved optimization algorithms or alternative formulations.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTS AND DETAILS

A.1 IMPACT OF ACTIVATION FUNCTIONS

We experimented with different activation functions to assess their impact on convergence and performance. Figure 3 shows the expert selection accuracy over epochs for None, ReLU, Leaky ReLU, and Sigmoid activations.

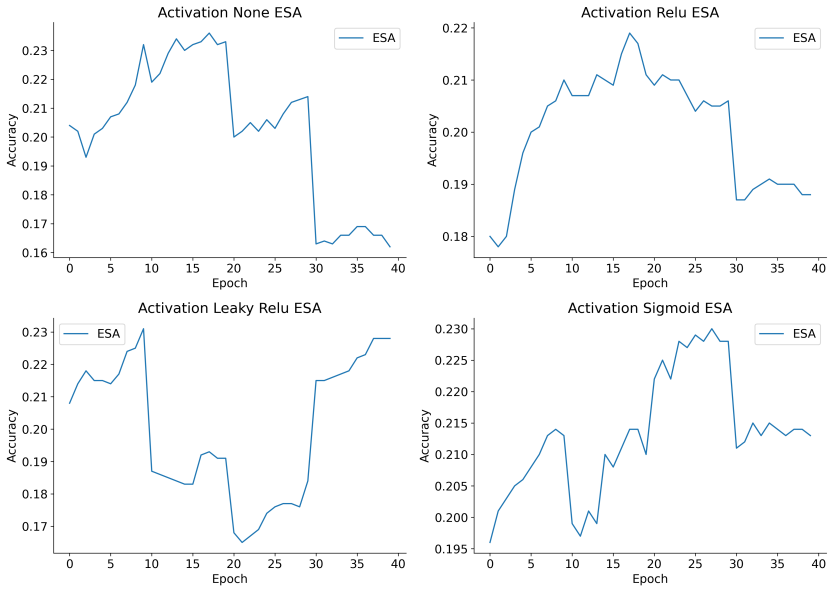

Figure 3: Expert Selection Accuracy over Epochs for different activation functions. Fluctuations persist across activations, indicating instability stems from the game-theoretic formulation.

The results indicate that while activation functions influence training dynamics, they do not fully address the convergence issues associated with higher q values.

A.2 EFFECT OF INITIALIZATIONS

The choice of weight initialization influenced the optimization process. Figure 4 compares training loss and expert selection accuracy for different initialization methods.

While some initializations led to marginally better curves, instability remained prevalent across methods.

A.3 DROPOUT IMPACT

We tested the impact of applying dropout regularization. Figure 5 shows training loss and accuracy curves with dropout rates of 20% and 50%.

High dropout rates led to degradation in performance, suggesting that excessive dropout exacerbates optimization difficulties.

A.4 IMPACT OF LEARNING RATE

We observed that the learning rate significantly affects performance. Figure 6 shows the final training loss and expert selection accuracy for different learning rates.

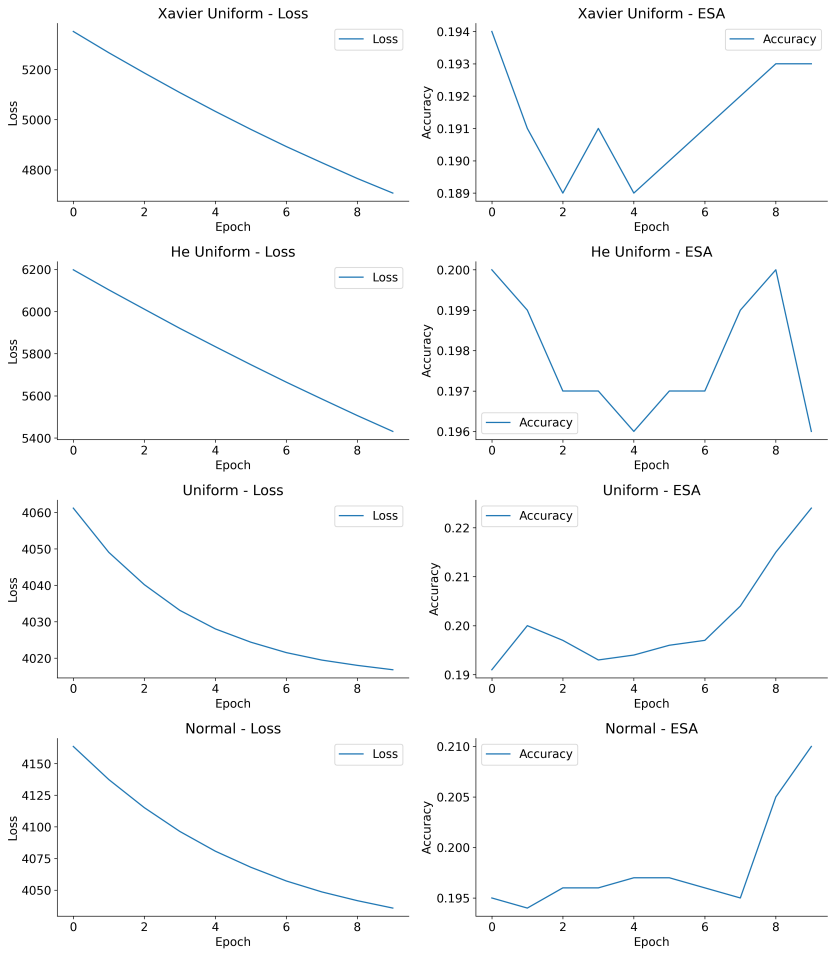

Figure 4: Training Loss and Expert Selection Accuracy with Different Weight Initializations. Appropriate initialization slightly improves convergence but doesn't resolve challenges.

Selecting an appropriate learning rate is crucial; however, even optimal rates don't fully resolve the instability.

B HYPERPARAMETERS

Table 2 lists the hyperparameters used in our experiments.

Table 2: Hyperparameters

Parameter	Value
Learning Rate	0.001, 0.01, 0.1
Batch Size	16, 64
Optimizers	Adam, SGD, RMSprop, AdamW
Weight Initialization	Xavier Uniform, He Uniform, Uniform, Normal
Activation Functions	None, ReLU, Leaky ReLU, Sigmoid
Epochs	50
Dropout Rates	0%, 20%, 50%

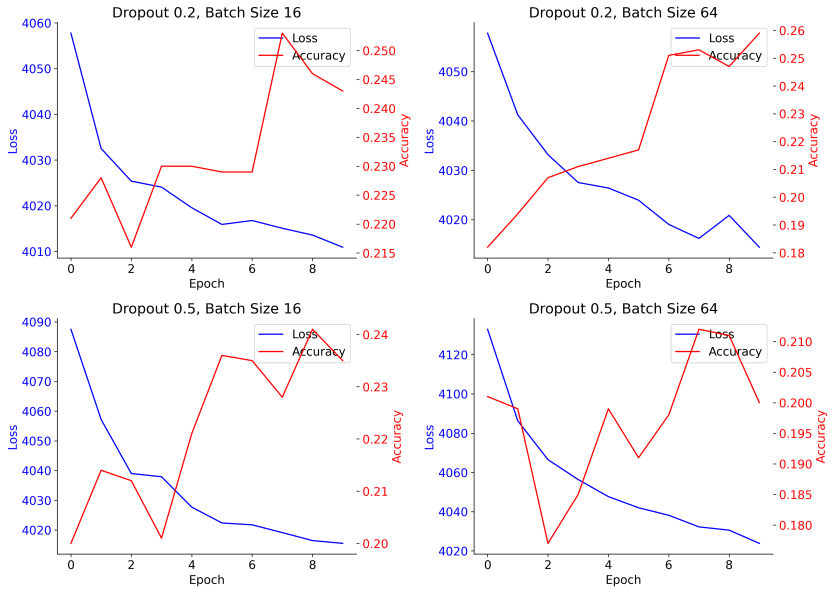

Figure 5: Training Loss and Accuracy with Different Dropout Rates. High dropout rates degrade performance, indicating regularization techniques need careful tuning.

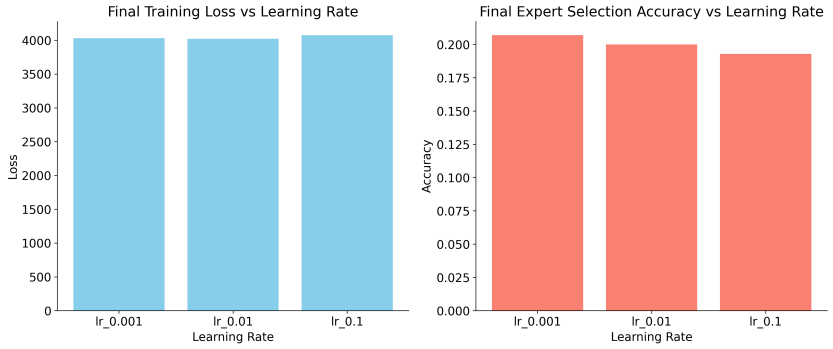

Figure 6: Final Training Loss (left) and Expert Selection Accuracy (right) for Different Learning Rates. Optimal learning rates do not fully resolve instability issues.